



Topic:

Swagger Framework — AI UI Test Generation + Self-Healing

Scope:

The Swagger Framework is the UI end-to-end sub-framework of the AI test platform. From a user story dropped into `inputs/swagger/`, it scrapes the live target page DOM via **Playwright MCP**, injects the story text into a base `swagger.yaml` description, and sends both to Anthropic Claude. Claude returns scenarios + manual test cases + a complete Playwright `.spec.ts` in a single call. A deterministic post-generation lint auto-fixes known anti-patterns (missing awaits, brittle waits, hard timeouts). Every locator in the generated spec is wrapped by the self-healing engine (SQLite cache + element-map + Claude fallback). If tests fail, an LLM patcher surgically rewrites only the failing test blocks and re-runs only those via Playwright's `--grep` filter. A 5-sheet Excel + Allure HTML report close the loop.

Key Features of the Swagger Framework:

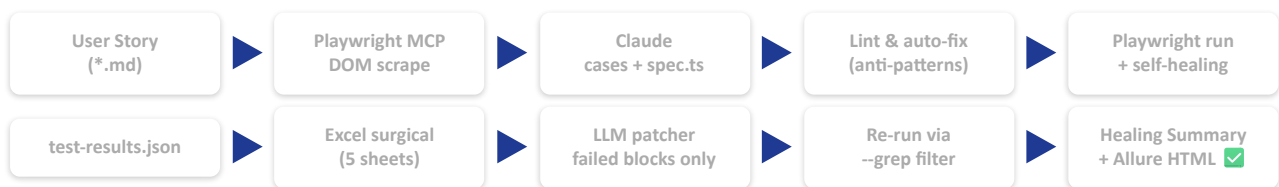
- **Live DOM Scrape via Playwright MCP:** Real CSS selectors are extracted from the actual page before generation — Claude sees what exists, not what it imagines.
- **One-Shot Generation:** A single Claude call returns scenarios, manual test cases AND the executable Playwright spec aligned 1-to-1 with the Excel rows.
- **Story Injection into swagger.yaml:** The orchestrator copies User Story text into `info.description` of the base OpenAPI spec before generation — story context follows the spec.
- **Deterministic Post-Gen Lint:** Auto-repairs known anti-patterns (missing `await`, hard `page.waitForTimeout`, fragile XPath) before saving the file.
- **Self-Healing Locators:** Every locator wrapped by the engine — 6-step priority ladder (SQLite cache → element-map → 3 heuristics → Claude). UI drift recovers at runtime, no source edits.
- **LLM Patch on Failure:** Failing test blocks rewritten surgically (only the failing `test()`), unaffected tests untouched, narrow re-run via `npx playwright test --grep "TC_001|TC_005"`.
- **5-Sheet Excel Report:** Test Scenarios, Test Cases, Defect Report, RTM, Healing Summary — surgical updates keep passing rows green across retries.
- **Allure HTML Report:** Per-test screenshots, video traces, console logs, network captures, healing events; opens automatically in browser.
- **Generic Target App:** Not hard-wired to OrangeHRM — change `BASE_URL` + drop a fresh story and the framework retargets.

Why Use the Swagger Framework?

UI tests rot fast — classes get renamed, wrappers get inserted, designers swap inputs for buttons. The Swagger Framework attacks this on three fronts. **At generation**, the LLM sees the real DOM via Playwright MCP, so it produces selectors that exist instead of plausible-looking guesses. **At runtime**, the self-healing wrapper recovers from drift transparently — the first failure caches a fix, every later run is a microsecond cache hit. **On failure**, the LLM patcher rewrites just the broken block — never the whole file — so a single failing assertion never costs the rest of the suite. Combined with surgical Excel updates and narrow Playwright re-runs, the result is a generated suite resilient enough to run in CI without daily human babysitting.

Swagger Framework Architecture

Three pipelines: Generation (story + DOM → Claude → spec), Execution (Playwright + self-healing), and Triage + Patch (LLM block rewrite → narrow re-run → surgical Excel).



Integration with Master Orchestrator, Tooling & CI/CD

Invoked via `node index.js --framework=swagger` (orchestrator) or `run-tests.bat` stand-alone. `swaggerAdapter.js` first injects the story into `swagger.yaml`, then runs the generator + Playwright. Built on **Node.js + Anthropic Claude + Playwright + Playwright MCP + TypeScript + ExcelJS + Allure**. Self-healing is the shared `self-healing/` module (sql.js + Claude). The orchestrator's `healingBridge.js` aggregates the per-run `healing-summary.json` into a consolidated view (`--healing-report`). Integrates with Jenkins / GitHub Actions / GitLab CI / Azure DevOps via the Playwright CLI + JSON reporter.

References: playwright.dev · github.com/microsoft/playwright-mcp · docs.anthropic.com · sql.js.org · docs.qameta.io/allure

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